GOOD MORNING!
早上好!
안녕하세요!

DAY 8-9

DAY 7 (FINAL PROJECT)

- 시스템 설계에 기반한 AMR 제어 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch 파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 8-9 (FINAL PROJECT)

- 개별 기능 통합 구현 및 Integration 테스트
- 통합 Launch 파일로 구현
- Robust한 시스템 구축을 위한 예외 처리 및 Code Refactoring
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 10 (FINAL PROJECT)

- 프로젝트 발표 및 시연
- 최종 산출문 정리(소스코드, 발표 PPT, 동작 영상)
- 팀 간 기술 컨퍼런스를 통한 기술 극복 경험담, 노하우 교류(채점 대상X)

프로젝트 RULE NUMBER ONE!!!

Have Fun Fun Fun!



PROJECT SCHEDULE REVIEW

BUSINESS/SOLUTION REQUIREMENT UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM REQUIREMENT UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

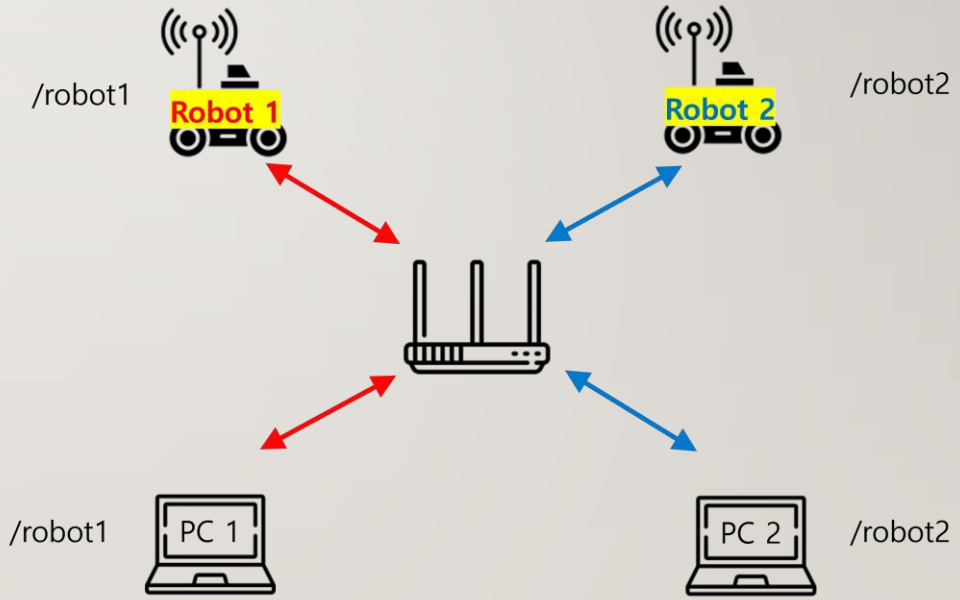
SYSTEM DESIGN ARCHITECTURE UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

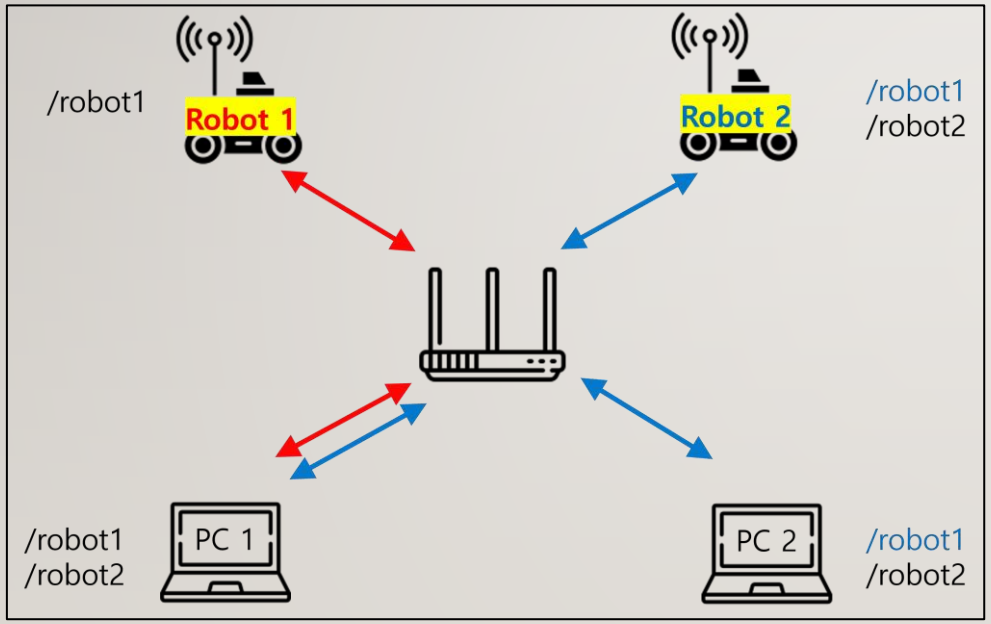
CHECK SOLUTION NETWORK SETTING

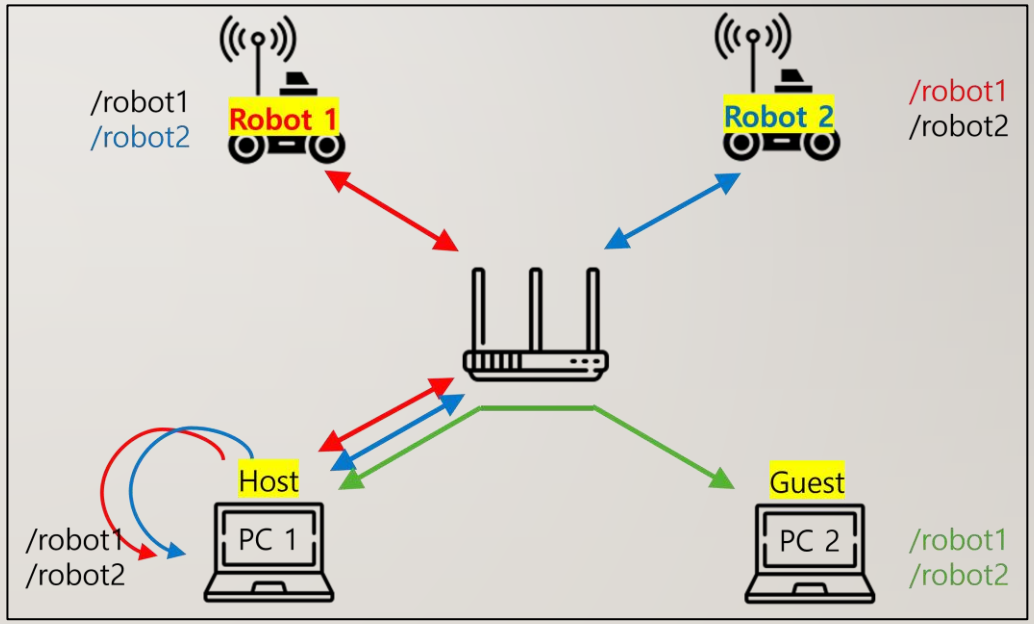
DAY 5 - System Monitor & Multi Robot

Aa 이름	상태
Multi Robot Standard Setup	시작 전
Multi Robot Custom Discovery Setup	시작 전



CHECK SOLUTION NETWORK SETTING



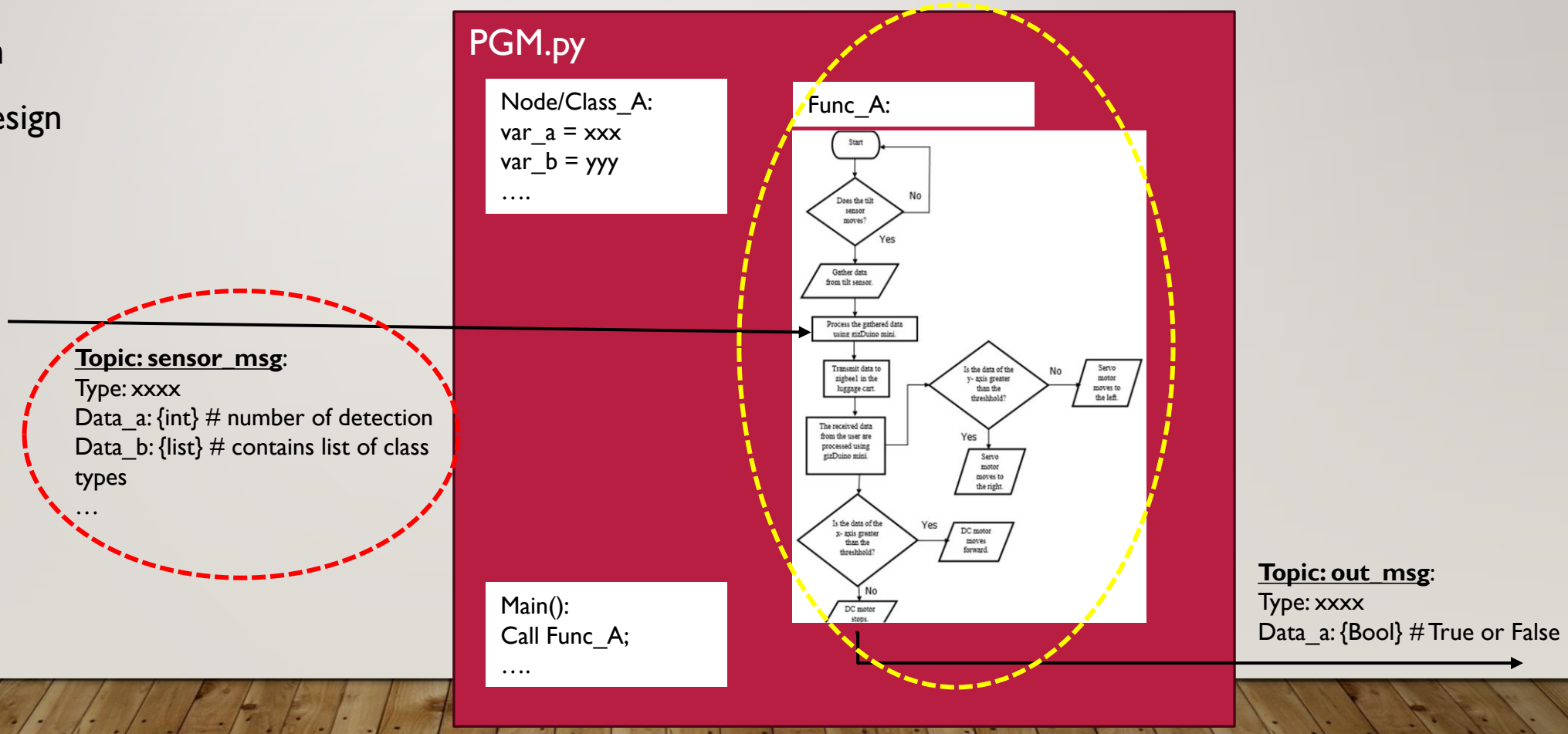


(평가 4) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

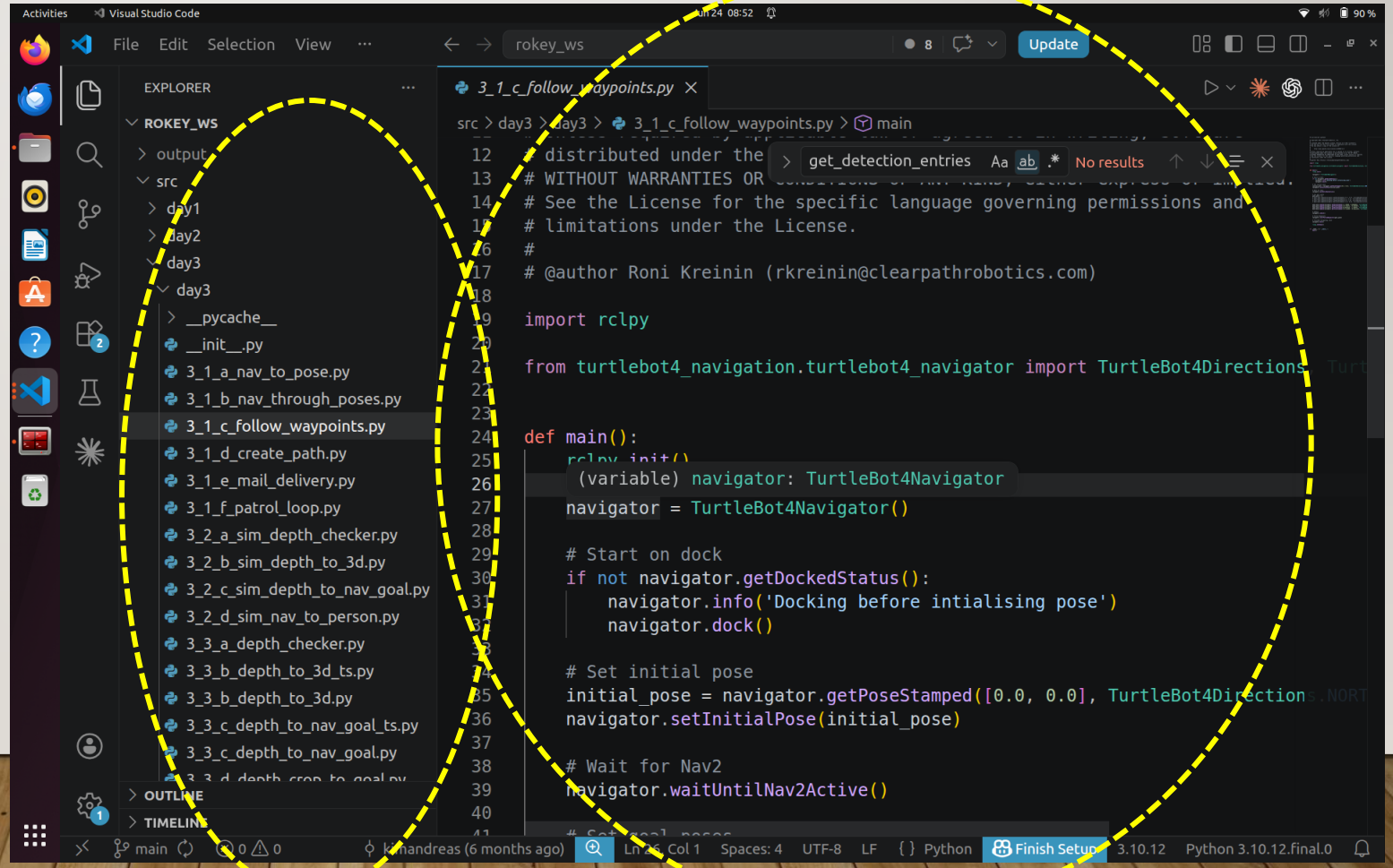
WHAT TO DO IN CODE REVIEW

- Explain
  - Design



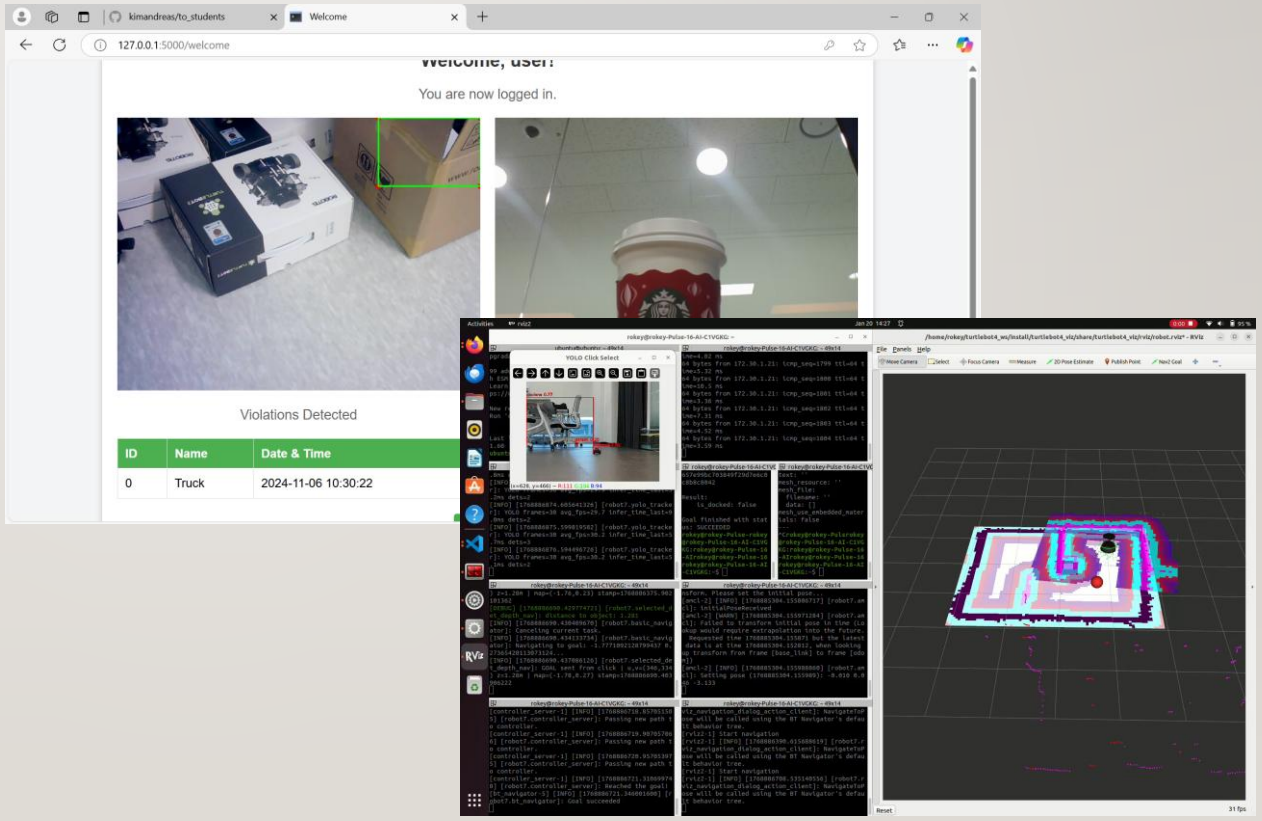
WHAT TO DO IN CODE REVIEW

- Explain
  - Design
  - Code
    - Must have #Comment's at Program, Node, Function, and in line (where appropriate) level
    - Including README.md would be good



WHAT TO DO IN CODE REVIEW

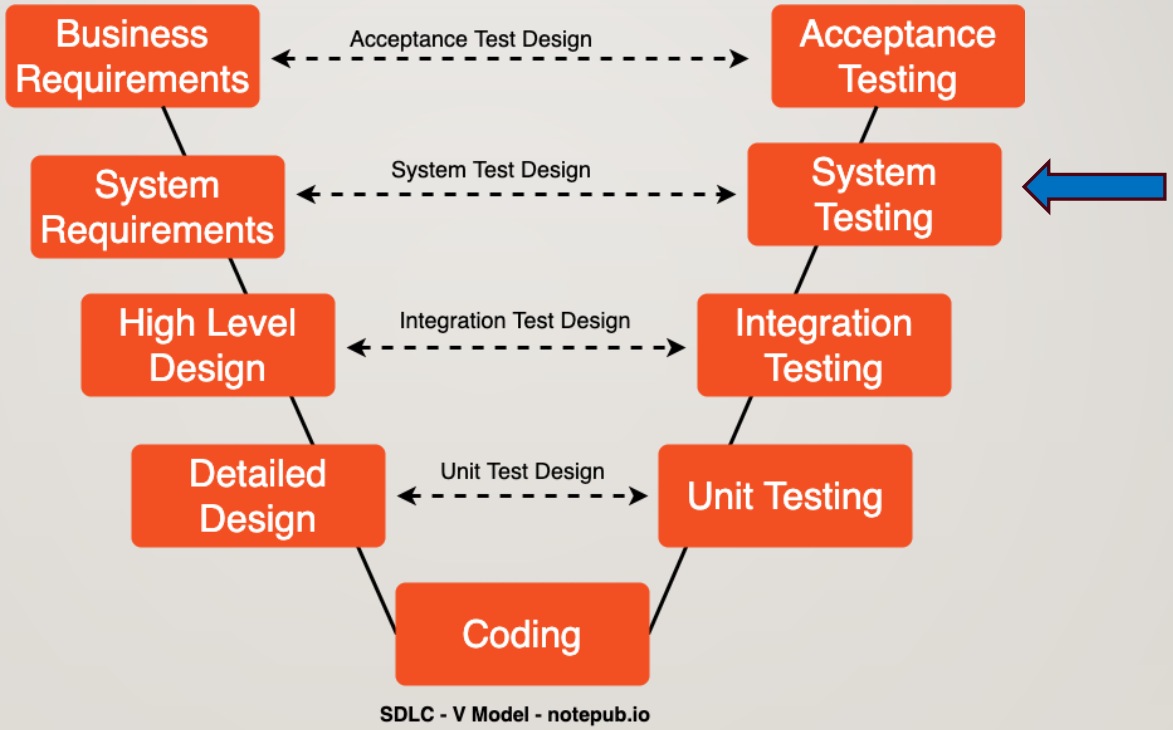
- Explain
  - Design
  - Code
  - Test/Validation
    - Test Code
    - Log Output
    - Screenshot/video
    - Photo/Video
    - ….



CODE REVIEW GUIDELINE

- No PPT!!!! Or demonstration
- Within 30 minutes
  - Describe your Design
  - Explain your Code
  - Show how your test and validated
  - Q&A
- One presenter with support from the team
- Come prepared in advance

SYSTEM INTEGRATION & TEST



TEAM EXERCISE 19

Perform system integration and test of All Modules

RESULTS & CODE REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

TEAM EXERCISE 20

Create System Design Document

Use the posted notes and flipchart as needed

EXAMPLE SYSTEM DESIGN DOCUMENT

- Project Overview
  - Summary of Project Requirement
  - System Requirement
- System Design Diagram (draw.io Attachment)
- Detail Description
  - Function
  - Dataflow
- Testing and Integration Plan



시스템 설계 문서 (SDD)

프로젝트 제목: 자율 이동 로봇(AMR) 보안 시스템
버전: 1.1
날짜: [날짜 삽입]

1. 개요

자율 이동 로봇(AMR) 보안 시스템은 단일 AI 기반 로봇을 사용하여 보안 구역 내에서 자율 순찰, 위협 탐지 및 경고를 제공하도록 설계되었습니다. 시스템은 단일 AMR 이 독립적으로 작동할 수 있도록 필요한 하드웨어 및 소프트웨어 구성 요소로 구성되며, 중앙 서버 없이 데이터를 현장에서 처리합니다.

2. 시스템 아키텍처

이 시스템은 단일 AMR 으로 구성되므로 데이터 처리, 네비게이션, 위협 탐지 및 경고가 모두 AMR 에서 로컬로 수행됩니다. AMR 은 모니터링, 알림 및 수동 제어를 위해 PC 의 사용자 인터페이스와 로컬 네트워크(Wi-Fi)를 통해 직접 통신합니다.

LAST DAY

- 9:30 – 11:00 p.m
  - System Integration & Test
  - Final Presentation Prep
- 11:30 – 12:30 p.m. ~ 2:15 – 4:15 p.m. (60 min/team)
  - Live/Video(one-take) Demonstration 5~10 minutes
  - Presentation ~30 minutes
  - Technical Sharing ~20 minutes
- 4:30 – 6:30 Equipment Return and Rap up

FINAL PRESENTATION MATERIAL PLANNING

- Solution Overview
- Key Issues and Challenges
  - How did you overcome
- Required Solution Improvements
- Lessons Learned
- Team Contribution

- Keep in under 10 slides (3 min/slide)

FINAL PRESENTATION MATERIAL PLANNING

- **[필수 공통사항]**
  - **과정명:** [두산로보틱스] 지능형 로보틱스 엔지니어
  - **팀 프로젝트명:** 예시) 삐용삐용 소방관
  - **주제:** 예시) (지능1) SLAM(위치추장 및 공간지도생성) 기반 자율주행 로봇 시스템 구현
  - **팀명:** 예시) A-1조 어벤져스
  - **팀원명:** 예시) 김두산, 이로키, 박캠프, 카리나(중도포기)
  - **멘토:** 예시) 앤디킴 강사님
  - 제출 파일은 반드시 **ppt -> pdf로 변환 필수**

* 역할 분담표에도 중도포기자 포함 (내용 없어도 무관)

SEND SYSTEM DESIGN DOC. HERE

산출물 구글드라이브

- https://drive.google.com/drive/folders/1AmPeoNG_nFoKaYIQh-1w9OTtoDhEkHTW?usp=sharing

Before: Day 10, 11:30 a.m.

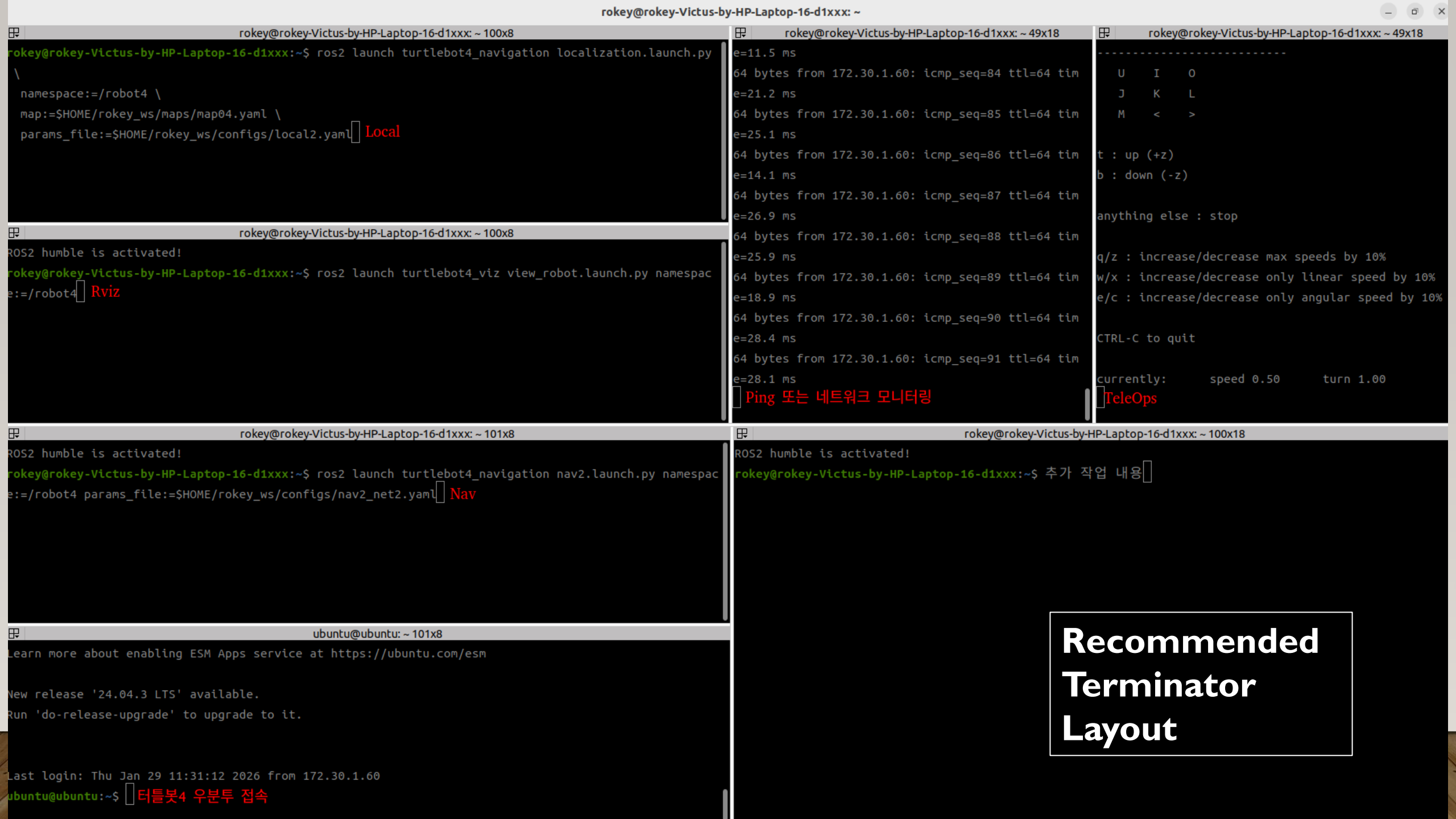
- ***Don't forget to include your Test and Integration Plan in the Doc.***

PROJECT DEMO 평가

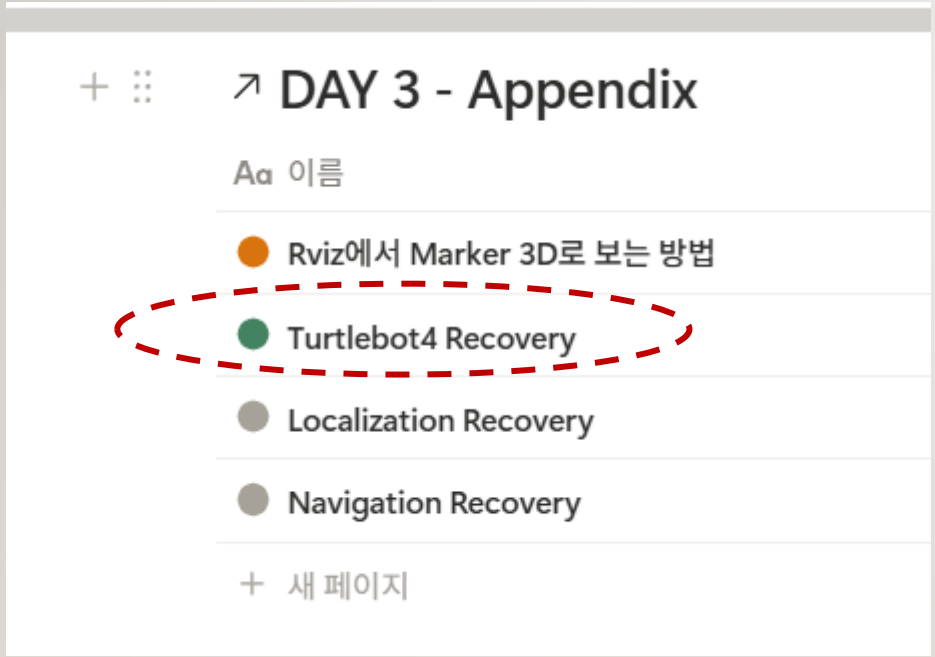
🔑평가 지표

항목	하위 항목	점수
1. 비즈니스 요구 사항 작성		10
2. Detection Alert Module의 코딩 및 테스트 수행		10
3. AMR Controller Module의 코딩 및 테스트 수행		10
4. System Monitor Module 코딩 및 테스트 수행		10
5. Process Flow Diagram을 사용하여 시스템 설계를 생성		30
6. System Design Doc 완성도		10
7. 모든 모듈의 시스템 통합 및 테스트 수행		
	Map 상 장애물 회피, Dock Undock init 자동화 goal arrival	10
	Detection: Object detection, depth 변환, 이동	10
	Realtime 장애/오류 Handling Function	10
	Quality of System Monitor Completed	10
	Level of System Integrated	20
8. 최종 프로젝트 발표		10
TOTAL		150

Map 상 장애물 회피, Dock Undock init 자동화 goal arrival
Detection: Object detection, depth 변환, 이동
Realtime 장애/오류 Handling Function
Quality of System Monitor Completed
Level of System Integrated


rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 100x8
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ ros2 launch turtlebot4_navigation localization.launch.py \
namespace:=/robot4 \
map:=$HOME/rokey_ws/maps/map04.yaml \
params_file:=$HOME/rokey_ws/configs/local2.yaml
Local
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 100x8
ROS2 humble is activated!
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ ros2 launch turtlebot4_viz view_robot.launch.py namespace:=/robot4
Rviz
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 49x18
e=11.5 ms
64 bytes from 172.30.1.60: icmp_seq=84 ttl=64 time=21.2 ms
64 bytes from 172.30.1.60: icmp_seq=85 ttl=64 time=25.1 ms
64 bytes from 172.30.1.60: icmp_seq=86 ttl=64 time=14.1 ms
64 bytes from 172.30.1.60: icmp_seq=87 ttl=64 time=26.9 ms
64 bytes from 172.30.1.60: icmp_seq=88 ttl=64 time=25.9 ms
64 bytes from 172.30.1.60: icmp_seq=89 ttl=64 time=18.9 ms
64 bytes from 172.30.1.60: icmp_seq=90 ttl=64 time=28.4 ms
64 bytes from 172.30.1.60: icmp_seq=91 ttl=64 time=28.1 ms
Ping 또는 네트워크 모니터링
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 49x18
U I O
J K L
M < >
t : up (+z)
b : down (-z)
anything else : stop
q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%
CTRL-C to quit
currently: speed 0.50 turn 1.00
TeleOps
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 101x8
ROS2 humble is activated!
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ ros2 launch turtlebot4_navigation nav2.launch.py namespace:=/robot4 params_file:=$HOME/rokey_ws/configs/nav2_net2.yaml
Nav
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx: ~ 100x18
ROS2 humble is activated!
rokey@rokey-Victus-by-HP-Laptop-16-d1xxx:~$ 추가 작업 내용
ubuntu@ubuntu: ~ 101x8
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Thu Jan 29 11:31:12 2026 from 172.30.1.60
ubuntu@ubuntu:~$
터틀봇4 우분투 접속
Recommended Terminator Layout

AMR RECOVERY



#checking the status of turtlebot4 bring up

- sudo systemctl status turtlebot4.service

- turtlebot4-source
- turtlebot4-service-restart
- turtlebor4-daemon-restart

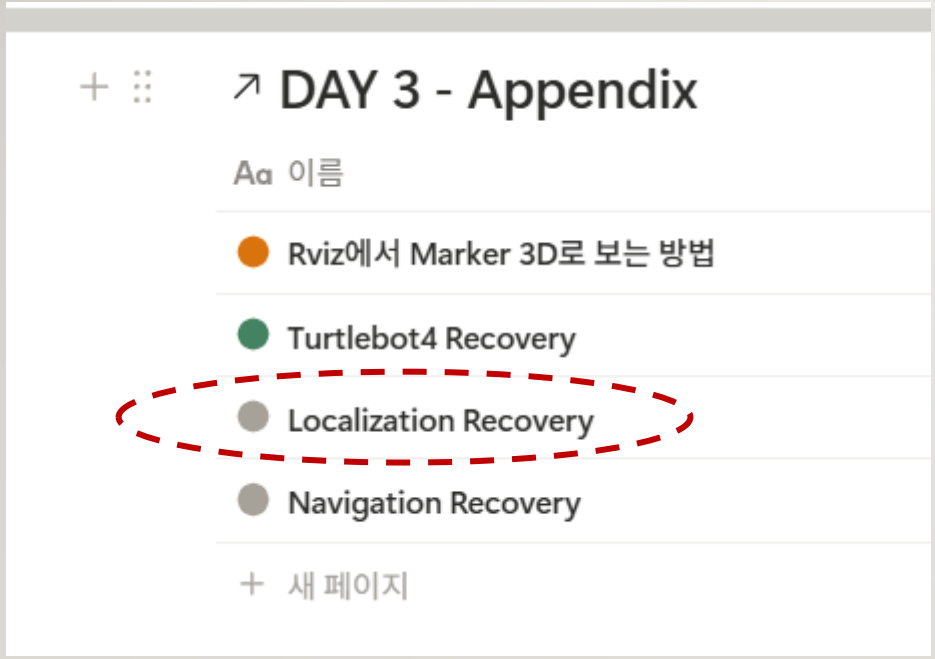
- sudo reboot

- sudo shutdown now

RECOVERY

- Localization should come up every time if AMR is stable
- Execute in the order of Localization, Rviz (set init_pose using 2d_pose_estimate; then undock), and Nav2
- If Nav2 is not coming up, then execute
  - ros-restart (ros2 daemon stop; ros2 daemon start), first
  - Then try executing navigation again

LOCALIZATION RECOVERY



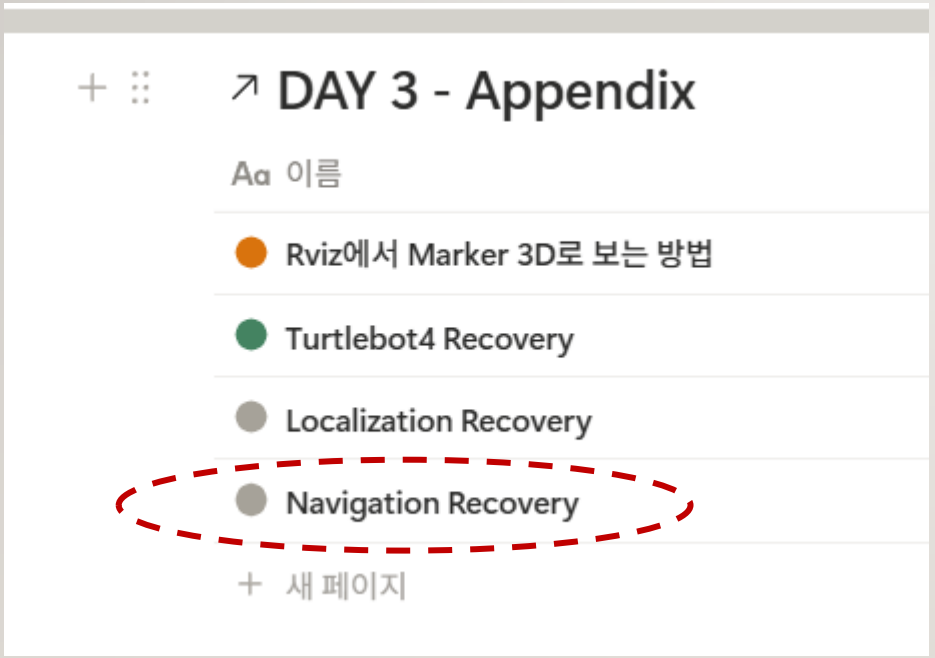
#When localization is not coming up

- $ ros2 lifecycle get /robot<n>/map_server
- $ ros2 lifecycle get /robot<n>/amcl

- $ ros2 lifecycle set /robot<n>/map_server configure
- $ ros2 lifecycle set /robot<n>/map_server activate

- $ ros2 lifecycle set /robot<n>/amcl configure
- $ ros2 lifecycle set /robot<n>/amcl activate

NAVIGATION RECOVERY




```
rokey@rokey-
rokey@rokey-P
        XX              X           XX      XXXXXX
         XXXXXX          XXXXXX     X XXXXX
              X                     X

ᕕ( ᐛ )ᕗ* 🌷 Nav2 Recovery Magic Starting!

========================================
      Nav2 Lifecycle Step List
========================================
  Step [1]: controller_server
  Step [2]: smoother_server
  Step [3]: planner_server
  Step [4]: behavior_server
  Step [5]: bt_navigator
  Step [6]: waypoint_follower
  Step [7]: velocity_smoother
========================================

로봇 네임스페이스 숫자 입력 (0-9): 8
실패한 Step 인덱스 (1-7): 1
----------------------------------------
현재 대상: robot8
복구 범위: controller_server 부터 마지막까지
----------------------------------------
[실행] /robot8/controller_server 설정 및 활성화 중...
Node not found
Node not found
[실행] /robot8/smoother_server 설정 및 활성화 중...
Node not found
Node not found
[실행] /robot8/planner_server 설정 및 활성화 중...
Node not found
```

프로젝트 RULE NUMBER ONE!!!

Are we having Fun???

